

AMMAR AYAZ

AI Engineer | GenAI Engineer | AI Automation | LLM & RAG Systems
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PROFESSIONAL SUMMARY

AI Engineer building production-grade agentic systems — multi-agent LangGraph orchestration, hybrid RAG pipelines, and on-premise LLM deployment. Shipped AIOps automation spanning 11 containerized services, a fully air-gapped finance assistant (zero external API calls, QLoRA fine-tuned Mistral 7B), and an 8-node CRAG research agent — all with FastAPI backends and OpenTelemetry tracing. B.S. Computer Science (AI focus), UET Peshawar — available October 2026.

TECHNICAL SKILLS

AI & Generative AI: LangGraph, LangChain, AI Agents, Multi-Agent Systems, RAG, Hybrid Retrieval, LLM Fine-Tuning (QLoRA), GPT-4o, Mistral, BGE Embeddings, Sentence-Transformers, RAGAS, MCP, Prompt Engineering
Backend & APIs: Python, FastAPI, REST APIs, WebSockets, PostgreSQL, Redis, Ollama, Model Serving
Vector DBs & Search: ChromaDB, FAISS, Pinecone, BM25, Cross-Encoder Reranking, Reciprocal Rank Fusion (RRF)
DevOps & Observability: Docker, Docker Compose, Kubernetes, GitHub Actions, CI/CD, OpenTelemetry, Prometheus, Loki, Jaeger, Grafana
ML & Data: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NLP, Deep Learning
Frontend: React, Next.js, TypeScript, JavaScript, Tailwind CSS, Streamlit

PROFESSIONAL EXPERIENCE

AI Engineer Intern — Agentic Systems & Automation

01/2026 – Present

MorpheLabs | Remote

- Design and deploy production AI automation workflows using GPT-4o, LangGraph, and FastAPI, enabling multi-step task execution across enterprise business processes.
- Develop and integrate AI agents with PostgreSQL, Redis, and external tools, supporting reliable data access, state management, and workflow orchestration.
- Instrument AI workflows end-to-end with OpenTelemetry-based distributed tracing, improving observability and accelerating debugging across production pipelines.
- Redesign agent orchestration and prompt evaluation pipelines, improving task-completion reliability and reducing automation failures across production environments.

PROJECTS

ASHIA — Agentic AIOps Self-Healing Infrastructure Platform | [GitHub](#)

LangGraph, FastAPI, React, Docker, PostgreSQL, Redis, Prometheus, Loki, Jaeger, ChromaDB

- Built an AI-driven AIOps platform with a 6-agent LangGraph pipeline spanning monitoring, root-cause analysis, remediation, and human-in-the-loop approval — automating end-to-end incident response from detection to resolution.
- Architected 11 containerized services orchestrated via Docker Compose, enabling isolated, independently scalable agents across the full incident lifecycle.
- Implemented Z-score anomaly detection across 12 Prometheus metrics with LLM-based root-cause analysis correlating Loki logs, Jaeger traces, and ChromaDB incident memory.
- Designed risk-tiered auto-remediation with human approval gates, eliminating manual triage for low-risk infrastructure incidents.

Sovereign On-Prem Finance AI Assistant | [GitHub](#)

Mistral 7B, QLoRA, Ollama, FAISS, BGE Embeddings, LangGraph, FastAPI, Docker, Streamlit

- Built a fully on-premise finance AI assistant with QLoRA fine-tuned Mistral 7B via Ollama — ensuring zero external API calls and complete data sovereignty across all inference.
- Engineered a 4-stage hybrid retrieval pipeline (BM25 → dense BGE embeddings → Reciprocal Rank Fusion → cross-encoder reranking), improving retrieval precision over single-method baselines.
- Designed a multi-step LangGraph agent with planning, sufficiency-checking router, AST-based calculator, and human-in-the-loop gate for reliable finance Q&A.
- Implemented PII redaction, hallucination self-check, and audit logging aligned to EU AI Act oversight requirements for compliant AI deployment.

AI Research Analyst — Agentic RAG Assistant | [GitHub](#)

LangGraph, ChromaDB, FastAPI, Streamlit, MCP, Docker

- Built an 8-node LangGraph research agent (route → retrieve → grade → rewrite → web-search → generate → grade-generation → respond) with CRAG-style adaptive document grading.
- Developed a multi-stage retrieval pipeline combining hybrid BM25 + dense ChromaDB search, query rewriting, and web-search fallback to maximize answer coverage and accuracy.
- Enforced answer groundedness at the response boundary — returning honest insufficient-evidence responses when retrieved documents fall below relevance threshold.
- Deployed via Docker Compose with an MCP server interface, redacted JSON execution traces, and full audit logging for transparent, inspectable operation.

EDUCATION

B.S. Computer Science — Artificial Intelligence Focus

10/2022 – 10/2026

University of Engineering & Technology (UET) Peshawar

CGPA: 3.4 / 4.0 | Relevant Coursework: AI, Machine Learning, Deep Learning, NLP, Algorithms, Databases

CERTIFICATIONS

DeepLearning.AI: LangChain for LLM Application Development • Building Systems with ChatGPT API • RAG for Production

Coursera: Machine Learning Specialization